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The Tesseract

In *Interstellar*, the entrance to the tesseract is a white checkerboard pattern. Each white square is the end of a beam. Cooper, entering the tesseract, falls down a channel between beams, dazed and confused, lashing out at what appear to be bricks along the channel wall, but turn out to be books. The channel leads to a large chamber, where he floats and struggles, gradually getting oriented.

The chamber is Christopher Nolan's unique take on one three-dimensional face of the four-dimensional tesseract, enhanced by Paul Franklin and his visual-effects team. The chamber and its environs are remarkably complex. Seeing them for the first time, I felt as disoriented as Cooper, even though I know what a tesseract is. Chris and Paul had enriched the tesseract so greatly that I only fully understood after talking with them.

Here's what I know—and what I learned, filtered through my physicist's eyes. I begin with the standard, simple tesseract, and then I build up to Chris's complexified tesseract.